AC circuits made visual

Read the circuit, calculate the current, then see why the phase angle has that sign.

01 KVL applies at every instant

Use passive reference polarities: current enters the positive terminal of each component.

$$v_s(t) = v_R(t) + v_L(t) + v_C(t)$$

02 Convert components into impedance

$\omega = 2\pi f$; $X_L = \omega L$; $X_C = 1/(\omega C)$. Use L in H and C in F.

$$Z = R + j(X_L - X_C)$$

03 Find magnitude and phase

$|Z| = \sqrt{[R^2 + (X_L - X_C)^2]}$; $\varphi = \tan^{-1}[(X_L - X_C)/R]$.

$$I = V_s / |Z|$$

04 Add voltage phasors

Taking current as the phase reference: $\bar{V}_s = V_R + j(V_L - V_C)$.

$$V_s = \sqrt{[V_R^2 + (V_L - V_C)^2]}$$

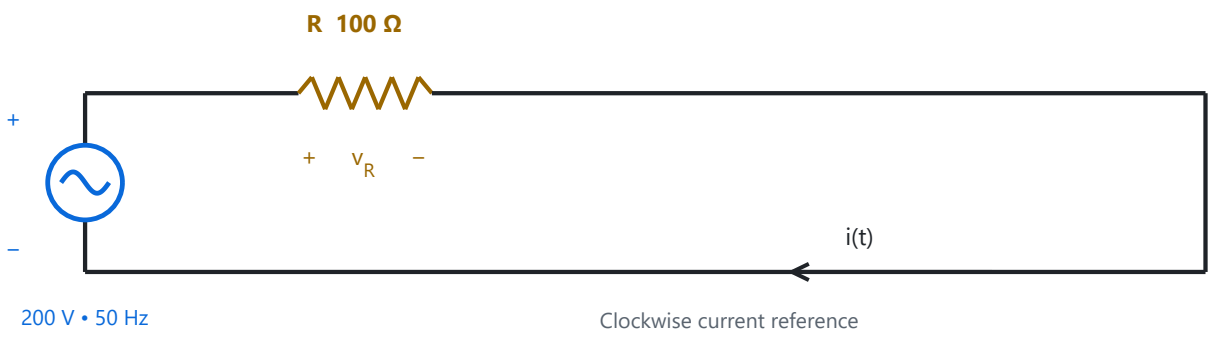
How to read the diagrams

Amber = resistor voltage; purple = inductor voltage; teal = capacitor voltage; blue = source voltage. Voltage arrow lengths share one scale within each diagram. The separate dashed current arrow shows direction only.

All examples assume ideal components and sinusoidal steady state. Transients after switching are excluded. Positive φ means current lags the source; negative φ means current leads.

01 Resistor only calculations

Find RMS current, impedance, phase angle, voltage drops and average power.



STEP 1 Write the circuit equation

$v_s(t) = Ri(t)$

STEP 2 Convert units and find reactance

For an ideal resistor, X = 0 and Z = R.

STEP 3 Find impedance magnitude

Z = 100 + j(0.00 − 0.00) = 100 + j0.00 Ω

$|Z| = \sqrt{[100^2 + (0.00)^2]} = 100.00\ \Omega$

STEP 4 Find current and phase

$I = V_s / |Z| = 200/100.00 = 2.000$ A

$\varphi = \tan^{-1}(0.00/100) = +0.00°$

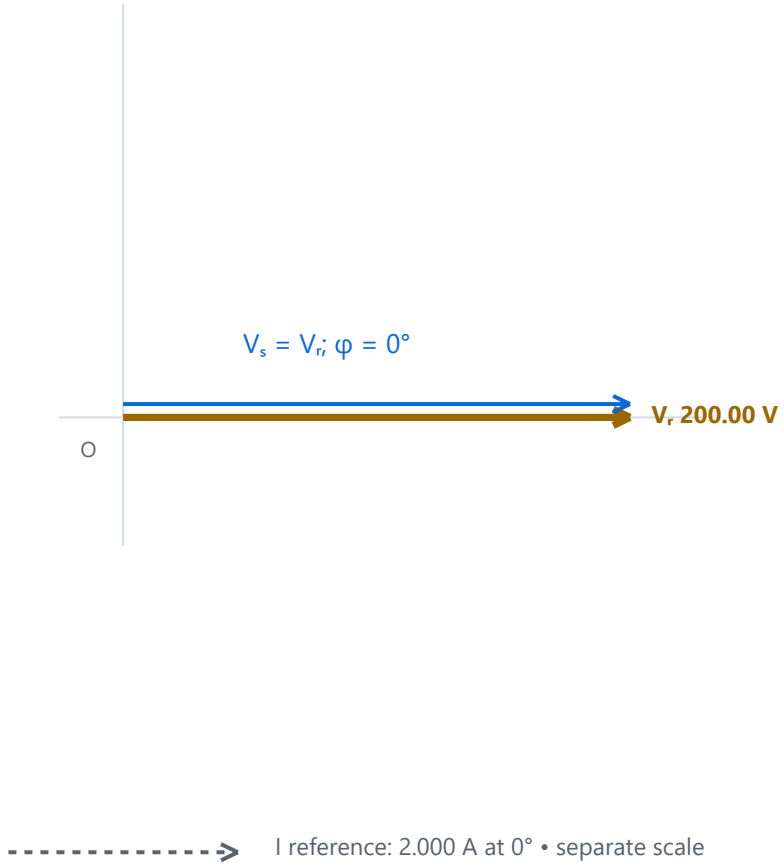
RESULT

Power factor = R/|Z| = 100/100.00 = 1.000 (in phase).

$I = 2.000$ A RMS $P = I^2R = 2.000^2 \times 100 \approx 400.0$ W

01 Resistor only phasors

Current is the 0° reference. Source voltage angle φ is measured from the positive horizontal axis.



CALCULATE THE VOLTAGE ARROWS

$V_R = IR = 2.000 \times 100 = 200.00$ V

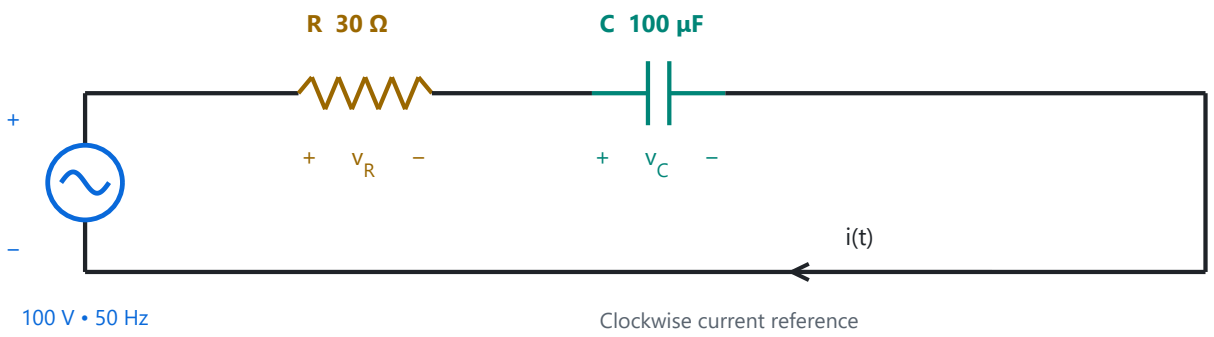
KVL CHECK

$\sqrt{[200.00^2 + (0.00 - 0.00)^2]} \approx 200$ V

The source and resistor voltages coincide. Voltage and current reach their peaks together.

02 Series RC calculations

Find RMS current, impedance, phase angle, voltage drops and average power.



STEP 1 Write the circuit equation

$v_s(t) = Ri(t) + v_C(t)$

STEP 2 Convert units and find reactance

$C = 100 \times 10^{-6}$ F; $X_C = 1/(2\pi \times 50 \times 100 \times 10^{-6}) = 31.83\ \Omega$

STEP 3 Find impedance magnitude

$Z = 30 + j(0.00 - 31.83) = 30 - j31.83\ \Omega$

$|Z| = \sqrt{[30^2 + (-31.83)^2]} = 43.74\ \Omega$

STEP 4 Find current and phase

$I = V_s / |Z| = 100/43.74 = 2.286$ A

$\varphi = \tan^{-1}(-31.83/30) = -46.70°$

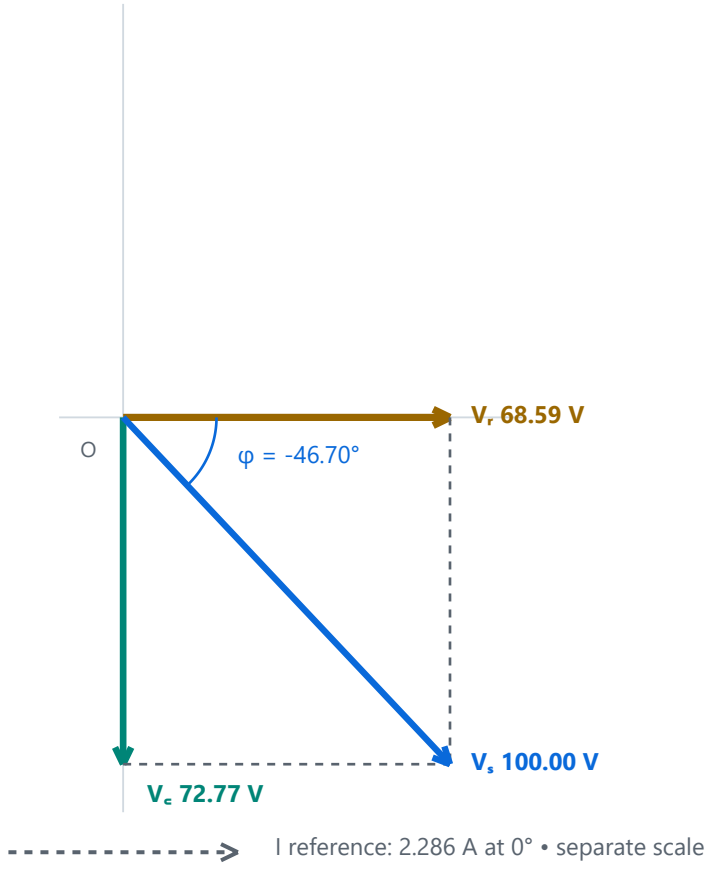
RESULT

Power factor = $R/|Z| = 30/43.74 = 0.686$ (leading).

$I = 2.286$ A RMS $P = I^2R = 2.286^2 \times 30 \approx 156.8$ W

02 Series RC phasors

Current is the 0° reference. Source voltage angle φ is measured from the positive horizontal axis.



CALCULATE THE VOLTAGE ARROWS

$V_R = IR = 2.286 \times 30 = 68.59$ V
$V_C = IX_C = 2.286 \times 31.83 = 72.77$ V

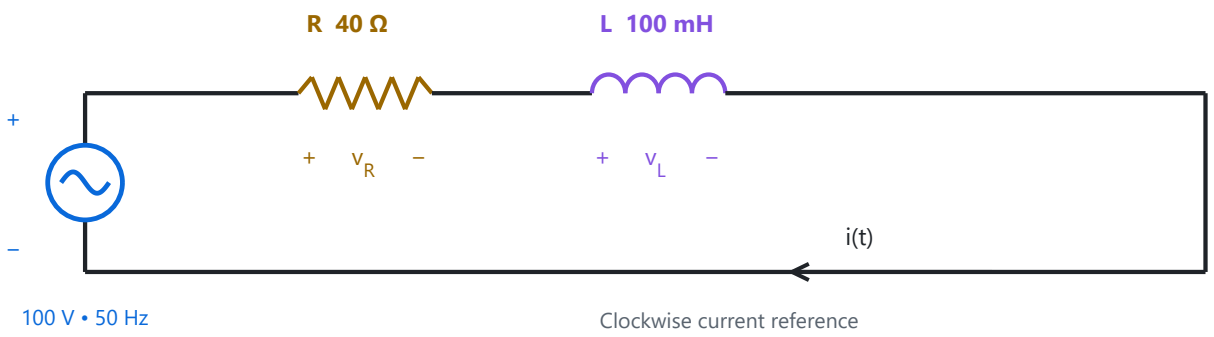
KVL CHECK

$\sqrt{[68.59^2 + (0.00 - 72.77)^2]} \approx 100$ V

The source phasor lies below the current reference. Its angle is negative, so current leads the source voltage.

03 Series RL calculations

Find RMS current, impedance, phase angle, voltage drops and average power.



STEP 1 Write the circuit equation

$v_s(t) = Ri(t) + L\,di/dt$

STEP 2 Convert units and find reactance

$L = 100 \times 10^{-3} = 0.10$ H; $X_L = 2\pi \times 50 \times 0.10 = 31.42\ \Omega$

STEP 3 Find impedance magnitude

$Z = 40 + j(31.42 - 0.00) = 40 + j31.42\ \Omega$

$|Z| = \sqrt{[40^2 + (31.42)^2]} = 50.86\ \Omega$

STEP 4 Find current and phase

$I = V_s / |Z| = 100/50.86 = 1.966$ A

$\varphi = \tan^{-1}(31.42/40) = +38.15°$

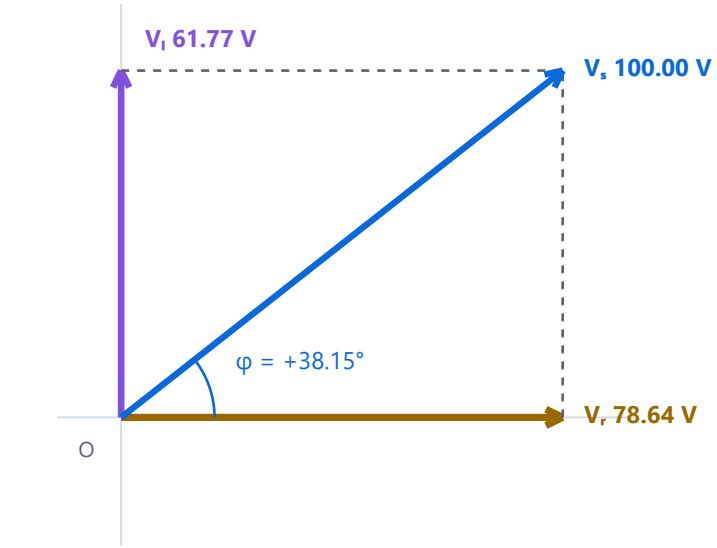
RESULT

Power factor = $R/|Z| = 40/50.86 = 0.786$ (lagging).

$I = 1.966$ A RMS $P = I^2R = 1.966^2 \times 40 \approx 154.6$ W

03 Series RL phasors

Current is the 0° reference. Source voltage angle φ is measured from the positive horizontal axis.



I reference: 1.966 A at 0° • separate scale

CALCULATE THE VOLTAGE ARROWS

$V_R = IR = 1.966 \times 40 = 78.64$ V
$V_L = IX_L = 1.966 \times 31.42 = 61.77$ V

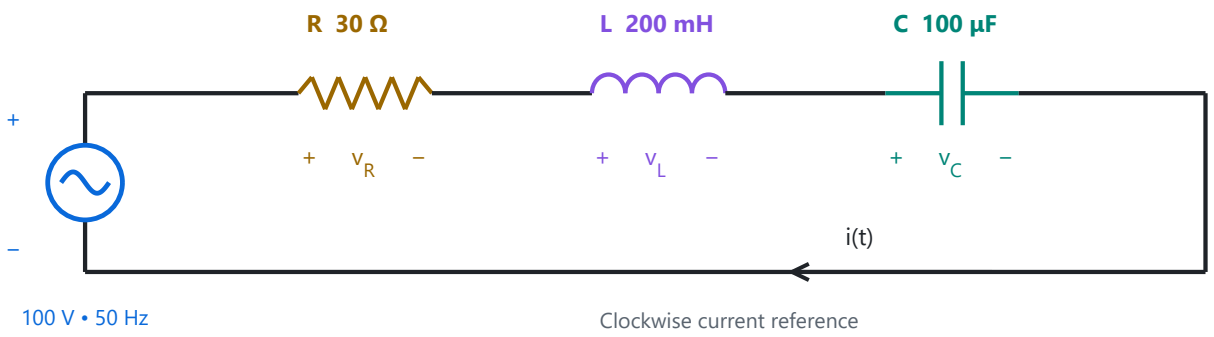
KVL CHECK

$\sqrt{[78.64^2 + (61.77 - 0.00)^2]} \approx 100$ V

The source phasor lies above the current reference. Its angle is positive, so current lags the source voltage.

04 Series RLC calculations

Find RMS current, impedance, phase angle, voltage drops and average power.



STEP 1 Write the circuit equation

$v_s(t) = Ri(t) + L\,di/dt + v_C(t)$

STEP 2 Convert units and find reactance

$L = 200 \times 10^{-3} = 0.20$ H; $X_L = 2\pi \times 50 \times 0.20 = 62.83\ \Omega$
$C = 100 \times 10^{-6}$ F; $X_C = 1/(2\pi \times 50 \times 100 \times 10^{-6}) = 31.83\ \Omega$

STEP 3 Find impedance magnitude

$Z = 30 + j(62.83 - 31.83) = 30 + j31.00\ \Omega$

$|Z| = \sqrt{[30^2 + (31.00)^2]} = 43.14\ \Omega$

STEP 4 Find current and phase

$I = V_s / |Z| = 100/43.14 = 2.318$ A
$\varphi = \tan^{-1}(31.00/30) = +45.94°$

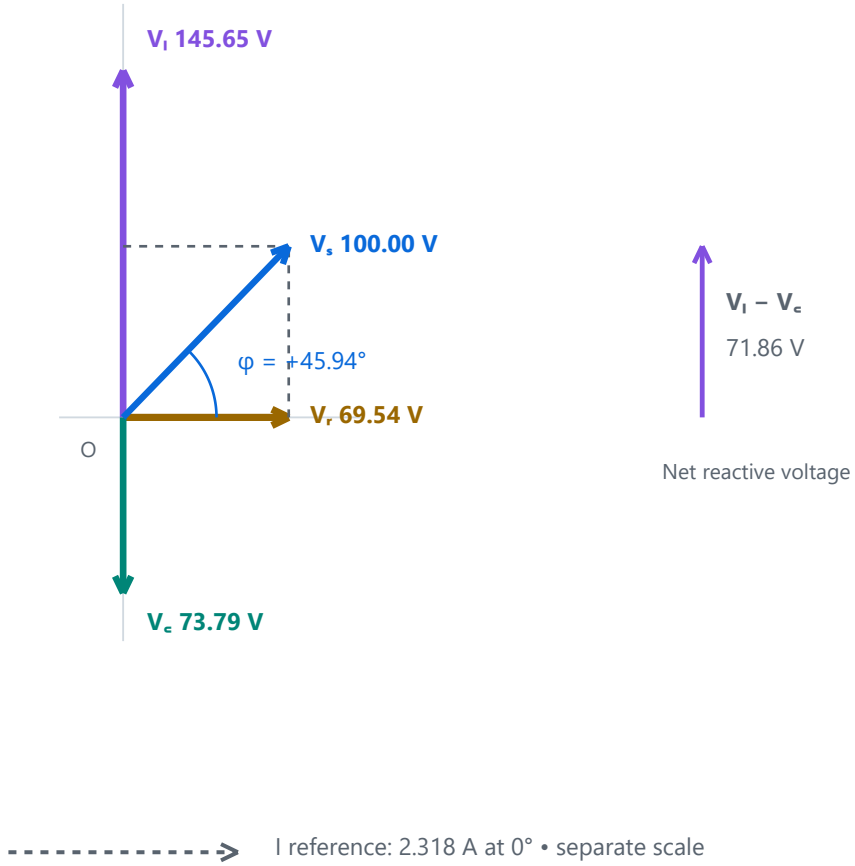
RESULT

Power factor = $R/|Z| = 30/43.14 = 0.695$ (lagging).

$I = 2.318$ A RMS $P = I^2R = 2.318^2 \times 30 \approx 161.2$ W

04 Series RLC phasors

Current is the 0° reference. Source voltage angle φ is measured from the positive horizontal axis.



CALCULATE THE VOLTAGE ARROWS

$V_R = IR = 2.318 \times 30 = 69.54$ V
$V_L = IX_L = 2.318 \times 62.83 = 145.65$ V
$V_C = IX_C = 2.318 \times 31.83 = 73.79$ V

KVL CHECK

$\sqrt{[69.54^2 + (145.65 - 73.79)^2]} \approx 100$ V

The upward inductor voltage exceeds the downward capacitor voltage. Their difference is upward, so current lags. Dashed lines show the resultant construction.

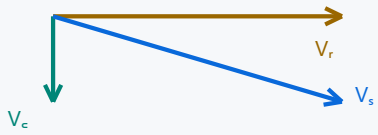
Compare the phase relationships

Qualitative diagrams below show direction only. Each row uses current as the horizontal reference.



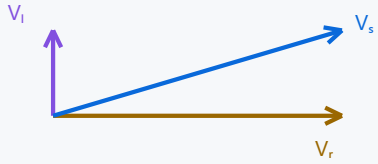
Resistor only

φ = 0°. Current and source are in phase.



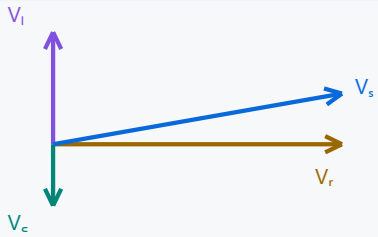
Capacitive circuit

φ < 0°. Current leads the source voltage.



Inductive circuit

φ > 0°. Current lags the source voltage.



RLC with XL > XC

Net voltage is upward; current lags.



Series resonance

XL = XC. Opposing reactive voltages cancel.

$f_0 = 1/(2\pi\sqrt{LC})$; $Z = R$; $I = V_s/R$.

Practice with working space

Questions 1 to 3. Show formula, substitution and units. Add a labeled phasor sketch.

01 R only

120 V, 60 Ω, 60 Hz. Find I, P and φ. Sketch the voltage phasor.



02 Series RC

100 V, 40 Ω, 100 µF, 50 Hz. Find |Z|, I and φ. Sketch the phasors.



03 Series RL

120 V, 30 Ω, 0.10 H, 50 Hz. Find |Z|, I, P and φ. Sketch the phasors.



Practice with working space

Questions 4 to 6. Show formula, substitution and units. Add a labeled phasor sketch.

04 Series RLC

100 V, 40 Ω, 0.10 H, 50 µF, 50 Hz. Find I and φ. Sketch the phasors.



05 Resonance

100 V, 20 Ω, 0.10 H, 100 µF. Find f_0 and current at resonance. Sketch the cancellation.



06 Explain KVL

Why can VL exceed the source voltage? Use a phasor sketch and the KVL magnitude equation.



Answers and phasor checks

Questions 1 to 3. Small rounding differences are normal. Mini diagrams are qualitative.

01 R only

I = 120/60 = 2.00 A; P = 2^2 × 60 = 240 W; φ = 0°.



Check: current reference is horizontal; source angle determines lead or lag.

02 Series RC

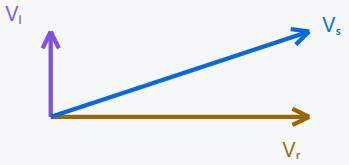
XC = 31.83 Ω; |Z| = 51.12 Ω; I = 1.956 A; φ = −38.51°.
Current leads.



Check: current reference is horizontal; source angle determines lead or lag.

03 Series RL

XL = 31.42 Ω; |Z| = 43.44 Ω; I = 2.762 A; P = 228.9 W; φ = +46.32°.



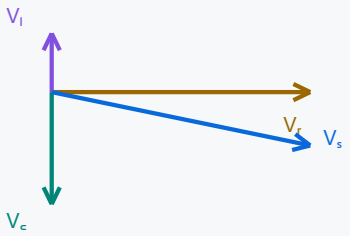
Check: current reference is horizontal; source angle determines lead or lag.

Answers and phasor checks

Questions 4 to 6. Small rounding differences are normal. Mini diagrams are qualitative.

04 Series RLC

X = −32.25 Ω; |Z| = 51.38 Ω; I = 1.946 A; φ = −38.87°.
Current leads.



Check: current reference is horizontal; source angle determines lead or lag.

05 Resonance

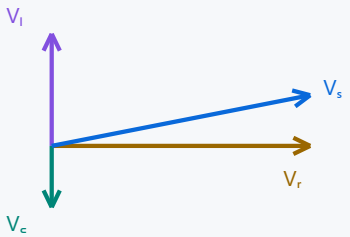
$f_0 = 1/(2\pi\sqrt{LC}) = 50.33$ Hz. At resonance, $I = V_s/R = 100/20 = 5.00$ A.



Check: current reference is horizontal; source angle determines lead or lag.

06 Explain KVL

VL and VC oppose each other. $V_s = \sqrt{[VR^2 + (VL - VC)^2]}$, so either reactive voltage may exceed V_s.



Check: current reference is horizontal; source angle determines lead or lag.

Parallel circuits made visual

RC, RL and RLC branches share one source voltage. Add currents using Kirchhoff's Current Law.

01 VOLTAGE IS SHARED

Each connected branch is directly across the same source.

$$V_R = V_L = V_C = V_s$$

02 CALCULATE EACH BRANCH CURRENT

Use RMS values. The resistor current is in phase with voltage.

$$I_R = V/R \quad I_L = V/X_L \quad I_C = V/X_C$$

03 CHOOSE VOLTAGE AS THE REFERENCE

I_R points right. I_C leads voltage by 90° and points up. I_L lags voltage by 90° and points down.

$$\bar{I}_s = I_R + j(I_C - I_L)$$

04 FIND MAGNITUDE AND CURRENT ANGLE

Positive θ: current leads voltage. Negative θ: current lags voltage.

$$|I_s| = \sqrt{[I_R^2 + (I_C - I_L)^2]}$$

$$\theta = \tan^{-1}[(I_C - I_L)/I_R]$$

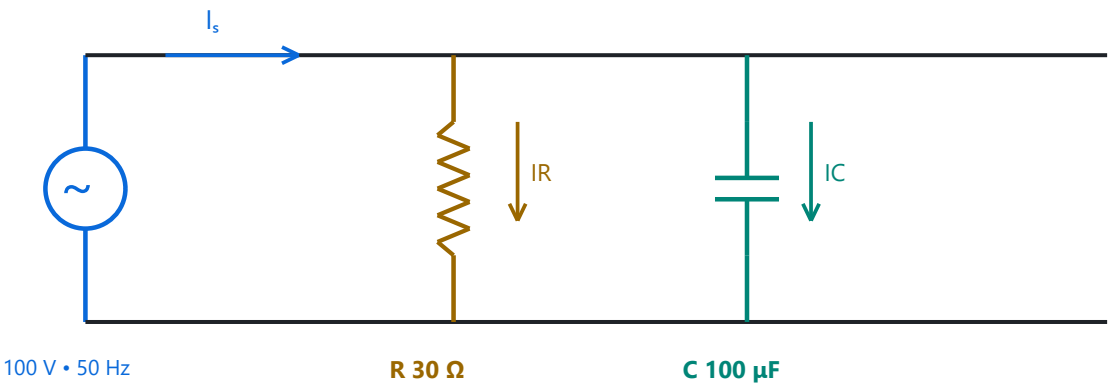
Do not mix the two angle conventions

The series pages use the impedance angle φ, measured from current to voltage. These parallel pages use current angle θ relative to voltage. They satisfy $\theta = -\varphi$.

For an omitted branch, its current is zero. All examples below use a 100 V RMS, 50 Hz source and ideal components.

Parallel RC calculations

Find the branch currents, total RMS source current, current angle, power and equivalent impedance.



STEP 1 SHARED VOLTAGE AND REACTANCE

Inductor branch omitted. $X_C = 1/(2\pi \times 50 \times 100 \times 10^{-6}) = 31.83\ \Omega$.

Every connected branch has V = 100 V RMS.

STEP 2 CALCULATE BRANCH CURRENTS

$I_R = V/R = 100/30 = 3.333$ A
$I_L = 0$ A (branch omitted)
$I_C = 100/31.83 = 3.142$ A

STEP 3 ADD PHASORS USING KCL

$\bar{I}_s = 3.333 + j3.142$ A

$|I_s| = \sqrt{[3.333^2 + (3.142)^2]} = 4.580$ A

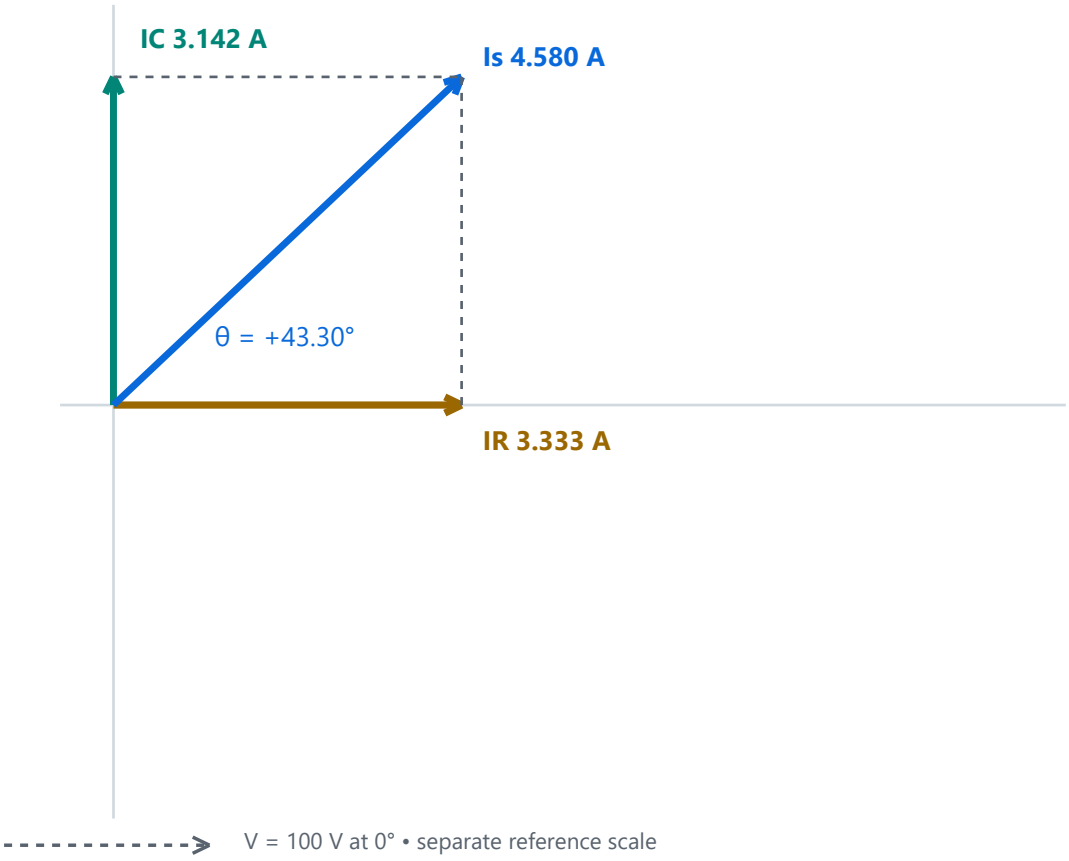
RESULT

$\theta = \tan^{-1}(3.142/3.333) = +43.30°$. Current leads voltage.
$|Z_{eq}| = V/|I_s| = 100/4.580 = 21.83\ \Omega$.

$P = V^2/R = 333.3$ W Power factor = 0.728

Parallel RC current phasors

Voltage is the horizontal 0° reference. All current arrows in this diagram use the same scale.



READ THE DIAGRAM

Capacitor current points up and exceeds inductor current. The total current leads voltage.

KCL AND POWER CHECK

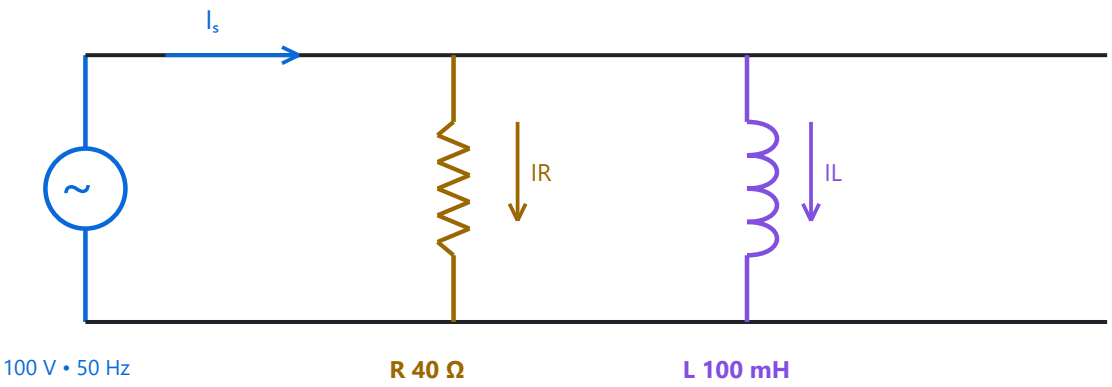
$I_R = 3.333$ A; $I_C - I_L = 3.142$ A.
$P = VI_s \cos\theta = 100 \times 4.580 \times 0.728 \approx 333.3$ W.

$$I_s = \sqrt{(I_R^2 + (I_C - I_L)^2)} = 4.580 \text{ A}$$

Compare with series: the same voltage is applied to every branch. Add the branch-current phasors; do not add current magnitudes directly.

Parallel RL calculations

Find the branch currents, total RMS source current, current angle, power and equivalent impedance.



STEP 1 SHARED VOLTAGE AND REACTANCE

$X_L = 2\pi \times 50 \times 0.10 = 31.42\ \Omega$. Capacitor branch omitted.

Every connected branch has V = 100 V RMS.

STEP 2 CALCULATE BRANCH CURRENTS

$I_R = V/R = 100/40 = 2.500$ A
$I_L = 100/31.42 = 3.183$ A
$I_C = 0$ A (branch omitted)

STEP 3 ADD PHASORS USING KCL

$\bar{I}_s = 2.500 - j3.183$ A

$$|I_s| = \sqrt{[2.500^2 + (-3.183)^2]} = 4.047\ \text{A}$$

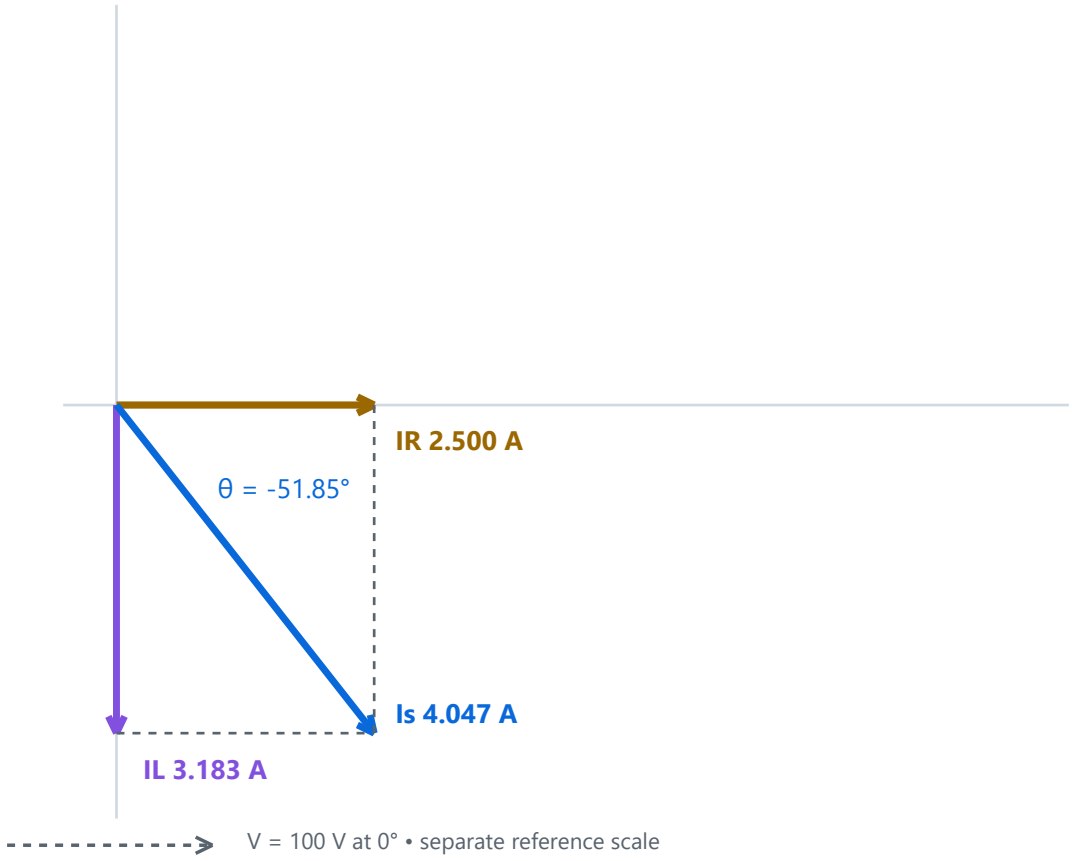
RESULT

$\theta = \tan^{-1}(-3.183/2.500) = -51.85°$. Current lags voltage.
$|Z_{eq}| = V/|I_s| = 100/4.047 = 24.71\ \Omega$.

$P = V^2/R = 250.0$ W Power factor = 0.618

Parallel RL current phasors

Voltage is the horizontal 0° reference. All current arrows in this diagram use the same scale.



READ THE DIAGRAM

Inductor current points down and exceeds capacitor current. The total current lags voltage.

KCL AND POWER CHECK

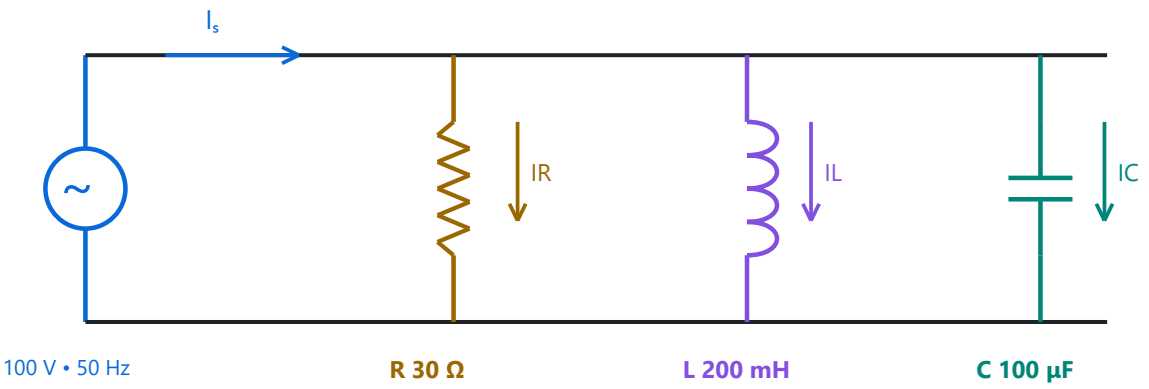
$I_R = 2.500$ A; $I_C - I_L = -3.183$ A.
$P = VI_s \cos\theta = 100 \times 4.047 \times 0.618 \approx 250.0$ W.

$$I_s = \sqrt{(I_R^2 + (I_C - I_L)^2)} = 4.047 \text{ A}$$

Compare with series: the same voltage is applied to every branch. Add the branch-current phasors; do not add current magnitudes directly.

Parallel RLC calculations

Find the branch currents, total RMS source current, current angle, power and equivalent impedance.



STEP 1 SHARED VOLTAGE AND REACTANCE

$X_L = 2\pi \times 50 \times 0.20 = 62.83\ \Omega$. $X_C = 1/(2\pi \times 50 \times 100 \times 10^{-6}) = 31.83\ \Omega$.

Every connected branch has V = 100 V RMS.

STEP 2 CALCULATE BRANCH CURRENTS

$I_R = V/R = 100/30 = 3.333$ A
$I_L = 100/62.83 = 1.592$ A
$I_C = 100/31.83 = 3.142$ A

STEP 3 ADD PHASORS USING KCL

$\bar{I}_s = 3.333 + j1.550$ A

$|I_s| = \sqrt{[3.333^2 + (1.550)^2]} = 3.676$ A

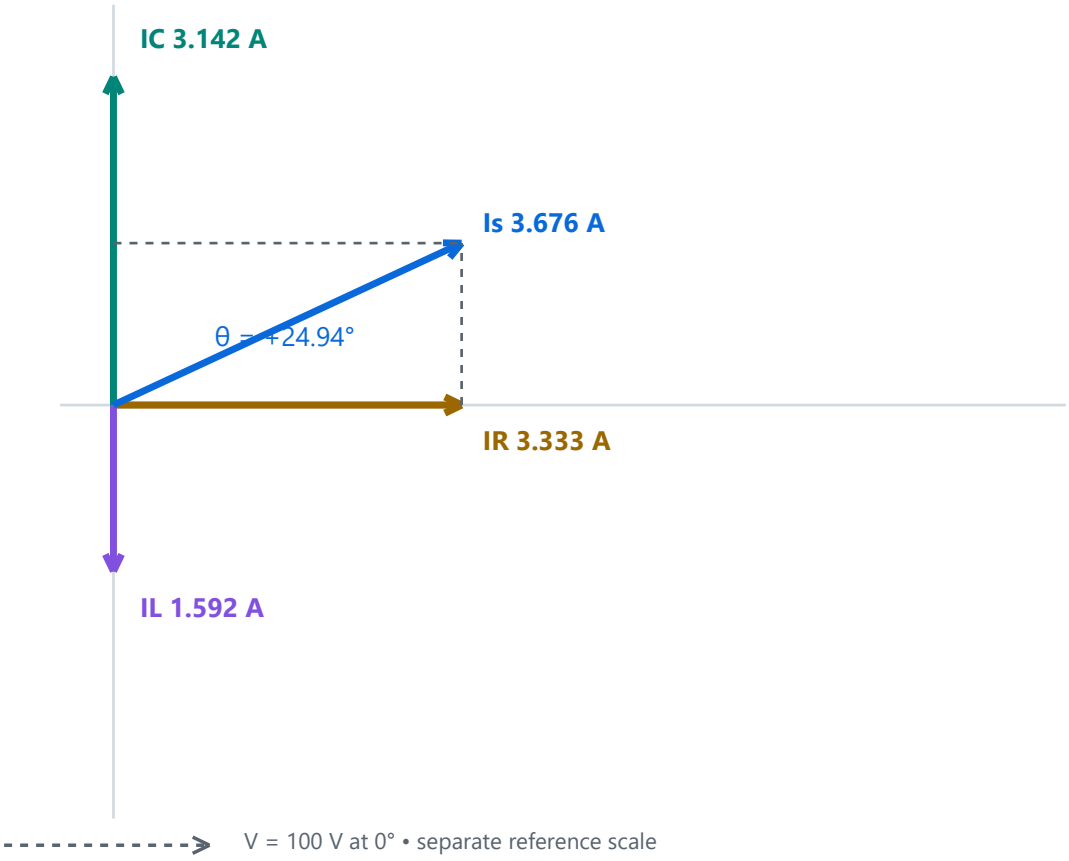
RESULT

$\theta = \tan^{-1}(1.550/3.333) = +24.94°$. Current leads voltage.
$|Z_{eq}| = V/|I_s| = 100/3.676 = 27.20\ \Omega$.

$P = V^2/R = 333.3$ W Power factor = 0.907

Parallel RLC current phasors

Voltage is the horizontal 0° reference. All current arrows in this diagram use the same scale.



READ THE DIAGRAM

Capacitor current points up and exceeds inductor current. The total current leads voltage.

KCL AND POWER CHECK

I_R = 3.333 A; $I_C - I_L$ = 1.550 A.
$P = VI_s \cos\theta = 100 \times 3.676 \times 0.907 \approx 333.3$ W.

$$I_s = \sqrt{(I_R^2 + (I_C - I_L)^2)} = 3.676 \text{ A}$$

At parallel resonance

$f_0 = 1/(2\pi\sqrt{LC})$ = 35.59 Hz. Then IL = IC, θ = 0°, and source current is minimum at V/R = 3.333 A. The individual branch currents remain nonzero.